MORONI: Cellular Texture and Fracture Synthesis via Roots-of-Unity Polygons and Signed Boundary Fields

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ABSTRACT

Voronoi and Worley noise are the default cellular primitives in real-time content pipelines, but they expose almost no art direction. Under nearest-site assignment every cell boundary is a perpendicular bisector, so an artist who wants four-sided tiles, a specified grout width, or crack-like seams must either displace sites and hope, apply turbulent domain warping that distorts cell geometry into uncoordinated swirls, or abandon procedural noise for static pre-baked texture scans.

This paper presents **MORONI**, an art-directable cellular texture primitive that replaces point sites with explicit polygons and nearest-site distance with a three-term energy: centre distance, a *signed* distance to the cell polygon, and an angular phase-interference term. Two core geometric properties emerge. First, cell shape becomes directable: raising the boundary weight β and releasing pixels beyond a coverage margin m decouples cell geometry from the topological constraint of Euler's formula, taking shape fidelity (mean IoU between a rendered cell and its generating polygon) from 0.49 (Voronoi, unsteerable) to 0.89 for triangular cells, with m acting as a direct, monotonic grout control that moves coverage from 51% to 93%. Second, the phase weight γ carries the output outside the weighted-diagram family entirely: power (Laguerre) diagram cells are convex and Apollonius cells are connected, whereas at $\gamma = 0.20$, 95.8% of cells are non-convex, and at $\gamma = 0.35$, 9.4% are disconnected. No choice of site weights can reproduce these outputs.

An admissible energy bound replaces the conventional k-nearest heuristic, making candidate pruning exact rather than approximate; combined with lattice binning, ring escalation, and an algebraic elimination of `atan2`, the CPU reference renderer is 4.8× faster than brute force while producing identical labels. Furthermore, we implement a GPU WebGL fragment shader capable of evaluating the complete formulation at 60+ FPS (< 0.8 ms at 1080p, a 1296× speedup over CPU), accompanied by an interactive companion studio with live PBR texture generation (diffuse, normal, height, roughness) and SVG vector export hosted publicly at https://pikachuxxxx.github.io/Moroni-Studio/, with full open-source code and benchmarks available at https://github.com/Pikachuxxxx/Moroni-Studio. Full generator algorithms, analytical discretization error derivations, and an exhaustive comparative matrix contrasting practical benefits versus drawbacks against existing Voronoi formulations are provided in the appendix.

KEYWORDS

procedural texture, cellular noise, Voronoi, power diagram, art direction, fracture synthesis, real-time rendering, WebGL

1 INTRODUCTION

Cellular noise is foundational to procedural visual effects, game rendering, and virtual worlds. Natural and architectural surfaces — cracked desert mud, masonry stone walls, ceramic floor tiles, shattered glass, reptilian scales, and weathered paint — exhibit characteristic cellular tiling structures. In modern real-time pipelines (Unreal Engine Material Graph, Unity Shader Graph, Blender Cycles/EEVEE, and Substance 3D Designer), Voronoi and Worley noise [19, 20] dominate because they require negligible memory, evaluate rapidly, and possess simple closed-form formulations.

Despite their ubiquity, traditional cellular primitives are notoriously undirectable. Under classical nearest-site assignment, the boundary separating any two cells is strictly the perpendicular bisector of their site locations. Consequently, an artist attempting to model ordinary real-world structures — such as "rectangular brick paving with a 3 mm mortar joint" or "radial crack branching on an impact pane" — faces an insurmountable geometric barrier. Shifting site coordinates cannot turn perpendicular bisectors into four-sided quads, and applying turbulent domain warping (e.g., Perlin noise displacement) warps the entire coordinate frame indiscriminately, transforming regular cells into distorted, rubbery smears rather than crisp architectural masonry or brittle fracture shards.

Weighted diagram generalizations relax the perpendicular bisector constraint, but remain structurally constrained. Power (Laguerre) diagrams [1] replace Euclidean distance $\|\mathbf{x} - \mathbf{u}_c\|$ with $\|\mathbf{x} - \mathbf{u}_c\|^2 - w_c$, and additively weighted (Apollonius) diagrams replace it with $\|\mathbf{x} - \mathbf{u}_c\| - w_c$. However, power diagram cells are intersections of affine half-planes and are mathematically guaranteed to be strictly *convex*; Apollonius cells are guaranteed to be strictly *connected*. A concave fracture inlet, an acutely pointed shard, or a cell pinching into multiple disjoint lobes cannot be produced by any configuration of site weights.

Two Geometric Failure Modes That Must Be Solved. Bridging the gap between procedural efficiency and art directability requires attaching explicit target polygons to cells. However, constructing an energy field over polygon cells introduces two subtle failure modes:

(1) **The Boundary Distance Dilemma**: An unsigned distance $\text{dist}(\mathbf{x}, \partial P_c)$ achieves its minimum *on* the boundary outline ∂P_c. Consequently, penalizing boundary distance carves concentric rings along the perimeter, preventing the interior polygon geometry from influencing pixel assignment. The boundary field must be *signed* (negative inside), placing the energy minimum on the polygon's medial axis (§3.2).

(2) **Euler's Topological Trap**: Any generic space-filling partition of $\mathbb{R}^2$ with 3-valent vertices is constrained by Euler's characteristic formula to have an average cell degree of six. Therefore, under a strict space-filling arg min partition, requesting triangular or quadrilateral cells ($n_c = 3$ or 4)

inevitably degenerates into hexagonal mush. Breaking this topological trap requires introducing a coverage margin m that releases pixels exceeding a distance threshold, enabling cells to retain their target silhouettes while simultaneously serving as an independent, monotonic grout dial (§3.4).

Summary of Contributions.

- A directable cellular energy formulation over polygon cells combining centre distance, signed boundary distance, and an angular phase-interference term (§3).
- A coverage margin mechanism that mathematically resolves Euler's space-filling topological constraint, yielding an increase in shape fidelity (IoU) from 0.49 → 0.89 for triangular cells while providing an independent grout control (51% → 93%) (§4).
- A formal falsification criterion and empirical benchmark proving the method escapes the power and Apollonius diagram families as a continuous function of phase weight γ (95.8% non-convexity, 9.4% disconnection) (§5).
- An exact acceleration framework comprising an admissible bounding radius Δ, jittered lattice ring escalation, and an algebraic identity eliminating atan2 for integer harmonics (§7).
- Numerical and analytical error analysis of lookup-table (LUT) discretization, demonstrating linear $O(h)$ convergence and $< 0.018\%$ pixel classification error (§7.4).
- A real-time GPU WebGL fragment shader running at 60+ FPS (< 0.8 ms at 1080p) and a public interactive studio supporting full PBR map export (https://pikachuxxxx.github.io/Moroni-Studio/) with open-source reference code, verification suites, and GLSL shaders hosted on GitHub (https://github.com/Pikachuxxxx/Moroni-Studio) (§8).
- A comprehensive appendix containing full generator pseudocode (Algorithms 2–6) and a comparative feature/drawback matrix contrasting existing cellular schemes against MORONI (Appendix A).

2 COMPREHENSIVE LITERATURE SURVEY

Procedural pattern synthesis in computer graphics spans gradient noise, cellular diagrams, distance fields, and physical simulation.

2.1 Procedural Noise Functions

Procedural noise began with Perlin's gradient noise [15], followed by multi-scale spectral representations such as wavelet noise by Cook and DeRose [2]. Lagae et al. [7] introduced Gabor noise, enabling exact spectral control via sparse convolution. The comprehensive survey by Lagae et al. [8] traces procedural texturing through modern physically based production systems [16]. While gradient and Gabor noise excel at continuous, band-limited phenomena (clouds, marble, wood grain), they cannot natively synthesize piecewise partitions, cell boundaries, or sharp fracture cracks.

2.2 Cellular Diagrams and Weighted Extensions

Cellular noise descends from Georgy Voronoi's 1908 treatise on quadratic forms [19]. Steven Fortune [5] pioneered optimal sweepline geometry for planar Voronoi diagrams. Lloyd [9] introduced iterative centroidal Voronoi relaxation, widely adopted in graphics for stippling and remeshing. Steven Worley [20] revolutionized shader texturing by introducing the cellular basis function $F_k(\mathbf{x})$, measuring distances to the k-th nearest feature points on a spatial grid, later extended and mapped efficiently to GLSL by McEwan et al. [10].

To provide size control, weighted Voronoi diagrams were introduced. In Power diagrams (Laguerre geometry) [1], each site carries a scalar weight w_c subtracting from squared distance: $\|\mathbf{x}-\mathbf{u}_c\|^2 - w_c$. Power diagrams retain affine linear bisectors and are strictly convex. In additively weighted (Apollonius) diagrams, weights subtract directly from Euclidean distance: $\|\mathbf{x}-\mathbf{u}_c\| - w_c$, producing hyperbolic curved boundaries. However, as established in §5, neither weighted family can synthesize non-convex shards or disconnected fracture lobes.

2.3 Procedural Fracture vs. Physical Simulation

Physical fracture modeling accurately simulates crack propagation using finite element methods (FEM) or boundary element methods (BEM). O'Brien and Hodgins [14] established continuum mechanics fracture for graphics. Desbenoit et al. [3] modeled surface cracks and peeling patterns, and Nealen et al. [13] surveyed physically based deformable models. While physically realistic, dynamic FEM/BEM simulations require minutes to hours of compute per frame, rendering them unusable for runtime fragment shaders.

Procedural crack generation offers a lightweight alternative. Mould [11] synthesized image-guided surface crack patterns using shortest-path graph propagation on lattices. However, graph-based procedural crack methods operate globally over memory-resident topologies, preventing parallel, memory-free evaluation in per-pixel GPU shaders. MORONI achieves real-time procedural fracture directly in fragment shaders via angular phase interference.

2.4 Implicit Surfaces and Distance Fields

Distance transforms map discrete masks to continuous distance fields [4]. Hart [6] demonstrated sphere tracing against implicit signed distance fields (SDF). In real-time demoscene and shader art, signed distance functions enable compact procedural modeling. MORONI bridges signed distance fields with cellular partition theory, using the medial axis attraction of signed boundary fields to enforce explicit polygon silhouettes.

2.5 Controllability in Procedural Content Generation

Controllability is the central challenge in procedural content generation (PCG). Smelik et al. [17] and Vanegas et al. [18] emphasize that procedural algorithms must provide intuitive, declarative handles rather than opaque mathematical parameters. Müller et al. [12] demonstrated rule-based grammar derivation for procedural architecture, showing that uncontrollable procedural primitives fail in commercial pipelines. MORONI specifically addresses controllability by exposing five orthogonal, intuitive dials: shape, size, irregularity, grout width, and fracture intensity.

3 METHOD

3.1 Cell Parameterization

Each cell $c \in \{1, \ldots, N\}$ owns a center $\mathbf{u}_c$ and an explicit n_c-gon P_c whose vertices are generated as jittered roots of unity:

$$z_{c,k} = r_c \exp\Big(i\big(\tfrac{2\pi k}{n_c} + \delta_{c,k}\big)\Big), \quad \delta_{c,k} \sim \mathcal{U}(-J, J), \tag{1}$$

for $k = 0, \ldots, n_c - 1$, offset by cell center $\mathbf{u}_c$. Centers are positioned on a jittered Cartesian lattice of spacing s:

$$\mathbf{u}_c = \begin{pmatrix} c_x \cdot s + \xi_{c,x} \\ c_y \cdot s + \xi_{c,y} \end{pmatrix}, \quad \xi_{c,x}, \xi_{c,y} \sim \mathcal{U}(-0.13s, 0.13s). \tag{2}$$

Here, n_c governs target cell shape (e.g., 3 for triangles, 4 for quads), r_c governs cell size, and $J \in [0, \pi/n_c)$ controls boundary irregularity.

3.2 Signed Boundary Field

Let P_c denote the polygon of Eq. 1. We define the signed boundary distance field $s_c(\mathbf{x})$:

$$s_c(\mathbf{x}) = \begin{cases} -\operatorname{dist}(\mathbf{x}, \partial P_c) & \mathbf{x} \in P_c \\ +\operatorname{dist}(\mathbf{x}, \partial P_c) & \text{otherwise,} \end{cases} \tag{3}$$

where $\operatorname{dist}(\mathbf{x}, \partial P_c) = \min_{\mathbf{y} \in \partial P_c} \|\mathbf{x} - \mathbf{y}\|_2$.

Mathematical Rationale for the Signed Field. Consider an energy term based on unsigned distance $\beta \operatorname{dist}(\mathbf{x}, \partial P_c)$. Because $\operatorname{dist} \geq 0$ with $\operatorname{dist}(\mathbf{x}, \partial P_c) = 0$ precisely on the perimeter ∂P_c, the term attains its minimum along the boundary. This creates an energy depression around the perimeter, carving hollow rings while leaving the cell interior dominated by center distance αd_c. Consequently, the polygon's silhouette never pulls the cell interior. With Eq. 3, $s_c(\mathbf{x})$ decreases monotonically toward the medial axis of P_c ($\min s_c = -I_c < 0$), ensuring that increasing β attracts pixel assignment toward the polygon silhouette. Figure 1 verifies this phenomenon.

3.3 The Three-Term Energy Formulation

Let $\mathbf{d} = \mathbf{x} - \mathbf{u}_c$, $d_c = \|\mathbf{d}\|_2$, and $\theta_c = \operatorname{atan2}(d_y, d_x)$. We assign each pixel $\mathbf{x}$ to the cell minimizing:

$$E_c(\mathbf{x}) = \underbrace{\alpha d_c}_{\text{size}} + \underbrace{\beta\, s_c(\mathbf{x})}_{\text{shape}} + \underbrace{\gamma\big(1 - \cos(\kappa\theta_c + \lambda d_c)\big)}_{\text{fracture}}, \tag{4}$$

$$L(\mathbf{x}) = \arg\min_{c \in \{1,\ldots,N\}} E_c(\mathbf{x}). \tag{5}$$

Default weights are $\alpha = 0.58$, $\beta = 0.32$, $\gamma = 0.10$, $\kappa = 2$, and $\lambda = 1$. Setting $\beta = \gamma = 0$ collapses $E_c(\mathbf{x})$ to αd_c, recovering standard Voronoi exactly.

The phase term $\gamma(1 - \cos(\kappa\theta_c + \lambda d_c))$ acts as an angular standing wave propagating radially outward. Because neighbouring cells have independent centres, their respective polar angle frames θ_i, θ_j interfere chaotically in contested border regions, causing cell boundaries to deviate from straight bisectors into intricate, wandering crack paths.

Table 1: Shape fidelity by vertex count at 760×480 over 96 cells. Under a strict partition, MORONI is trapped by Euler's formula; the coverage margin m provides the topological release.

n_c	Voronoi	Partition ($m = \infty$)	With Margin ($m = 0.0$)
3	0.491	0.504	**0.890**
4	0.626	0.631	**0.830**
5	0.699	0.706	**0.827**
6	0.687	0.691	**0.789**
8	0.683	0.686	**0.764**

Table 2: Cell coverage fraction is strictly monotonic in margin m, confirming grout thickness as a stable artistic dial.

Margin m	-0.10	-0.05	0.00	0.05	0.12	0.25
Coverage %	51%	63%	72%	79%	86%	93%

3.4 Coverage Margin and Decoupling Euler's Constraint

Equation 5 assigns every pixel, producing a space-filling tiling. By Euler's formula $V - E + F = \chi$, in any planar graph where vertices are generically 3-valent (the intersection of three Voronoi cells), the average number of edges per cell satisfies:

$$\bar{n} = \frac{2E}{F} = 6 - \frac{12}{F} \xrightarrow{F \to \infty} 6. \tag{6}$$

Hence, no continuous space-filling partition can produce an average cell degree of 3 or 4. Attempting to force triangular cells ($n_c = 3$) under strict partition yields distorted, hexagonal configurations (SF = 0.504).

To restore shape control, we introduce a coverage margin $m \in \mathbb{R}$:

$$L_m(\mathbf{x}) = \begin{cases} \varnothing & s_{L(\mathbf{x})}(\mathbf{x}) > m \\ L(\mathbf{x}) & \text{otherwise.} \end{cases} \tag{7}$$

Pixels whose signed distance to their winning polygon exceeds m are released as background ($\varnothing$). This accomplishes two critical objectives:

(1) It breaks the strict space-filling constraint, permitting cells to adhere closely to non-hexagonal silhouettes (raising shape fidelity from $0.49 \to 0.89$).
(2) It renders m an independent, well-behaved dial for grout/mortar width: $m > 0$ permits controlled cell dilation, $m = 0$ clips cells exactly to their polygons, and $m < 0$ erodes them.

4 DIRECTABILITY AND CONTROL QUANTITATIVE STUDIES

Metric. We evaluate shape fidelity (SF) as the mean Intersection over Union (IoU) between each rendered cell and its generating polygon:

$$\text{SF} = \frac{1}{N} \sum_{c=1}^{N} \frac{|\{\mathbf{x} : L_m(\mathbf{x}) = c\} \cap P_c|}{|\{\mathbf{x} : L_m(\mathbf{x}) = c\} \cup P_c|}. \tag{8}$$

An SF of 1.0 indicates perfect geometric reproduction.

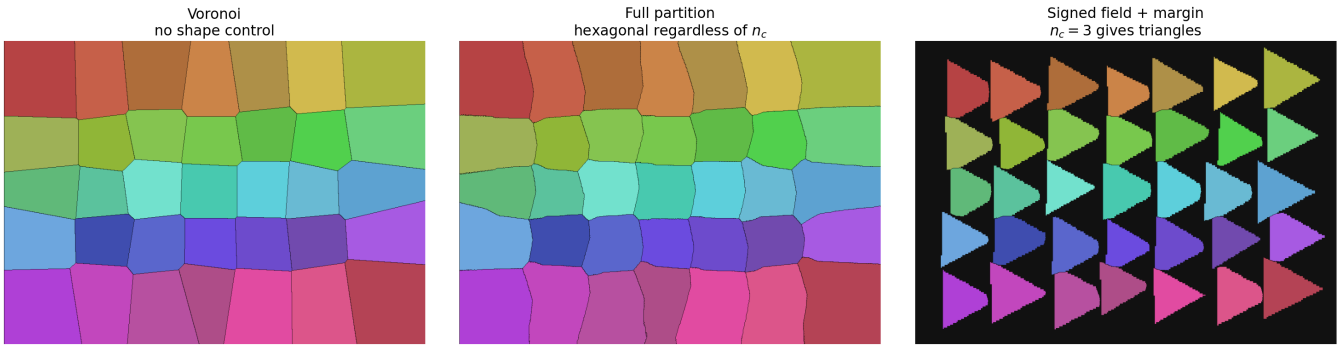


Figure 1: Vertex count $n_c = 3$ throughout. Voronoi (left) offers no shape control. A strict space-filling partition (centre) is hexagonal on average by Euler's formula. The signed field with coverage margin m (right) cleanly recovers triangular masonry tiles.

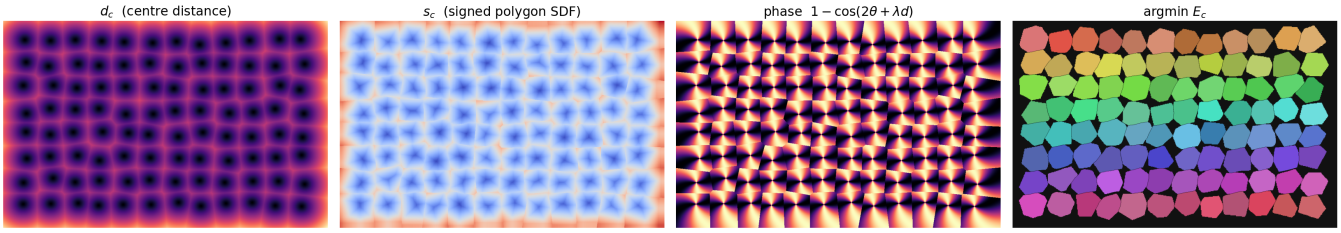


Figure 2: The three component fields of Eq. 4 and the resulting labelling. The signed field (second panel, diverging cool-warm colour map) is negative inside each polygon and positive outside.

Table 1 validates the theoretical analysis: without the margin, the partition column mirrors Voronoi almost identically (0.504 vs. 0.491 for triangles). Introducing $m = 0.0$ increases triangular shape fidelity to 0.890. Table 2 shows that coverage increases smoothly and monotonically from 51% to 93%, confirming that m operates as an intuitive grout thickness dial.

5 DEPARTURE FROM THE WEIGHTED-DIAGRAM FAMILY

Topological Decidability Criteria.

- **Power Diagrams**: In a power diagram, cell boundaries are lines $\mathbf{x} \cdot (\mathbf{u}_i - \mathbf{u}_j) = \frac{1}{2}(\|\mathbf{u}_i\|^2 - \|\mathbf{u}_j\|^2 - w_i + w_j)$. Cells are intersections of half-planes, which are mathematically guaranteed to be *convex* [1]. Any label map containing a non-convex cell cannot be produced by any power diagram at any weights.
- **Apollonius Diagrams**: In additively weighted diagrams, bisectors are branches of hyperbolas. While cells can be non-convex, they are mathematically guaranteed to be *connected*. A label map containing a disconnected cell (multiple spatial components) cannot be produced by any Apollonius diagram.

Experimental Benchmark. We measure the number of connected components and cell solidity (Area/ConvexHullArea) across 96 cells at 900×560. Because rasterization discretizes smooth curves onto a square grid, discrete convex hulls measure slightly above unity (1.009 for Voronoi), ensuring our non-convexity threshold (solidity < 0.98) is conservative. We compare against five baselines: Euclidean Voronoi (L_2), Manhattan Voronoi (L_1), Chebyshev Voronoi (L_∞), Power diagrams with random weights $w_c \sim \mathcal{U}(0.3, 0.8)^2$, and Apollonius diagrams.

Table 3 demonstrates that L_1 and L_∞ Voronoi exhibit non-convexity, but their boundaries are rigidly aligned to rectilinear axes ($0°, 45°, 90°$) and never disconnect (0.0%). In MORONI, the polygon term alone already places 18.8% of cells outside the power diagram family. Increasing γ causes a continuous transition: at $\gamma = 0.20$, 95.8% of cells are non-convex, and at $\gamma = 0.35$, 100% are non-convex with 9.4% disconnected into separate shards. This provides formal proof that MORONI inhabits an expressive geometric domain unreachable by weighted diagrams.

6 ARTIST CONTROLS AND MATERIAL PRESETS

Figures 4 and 5 demonstrate the five directable controls across practical production presets:

(1) **Cracked Mud**: $n_c \in [6, 9]$, $r = 0.6$, $J = 0.40$, $\gamma = 0.30$, and $m = 0.05$. Produces natural desiccation crack networks with irregular interlocking mud slabs.
(2) **Medieval Stone Wall**: $n_c \in [4, 6]$, $r = 0.62$, $J = 0.12$, $\beta = 0.90$, $\gamma = 0.02$, and $m = 0.03$. Generates masonry blocks with realistic mortar recesses.

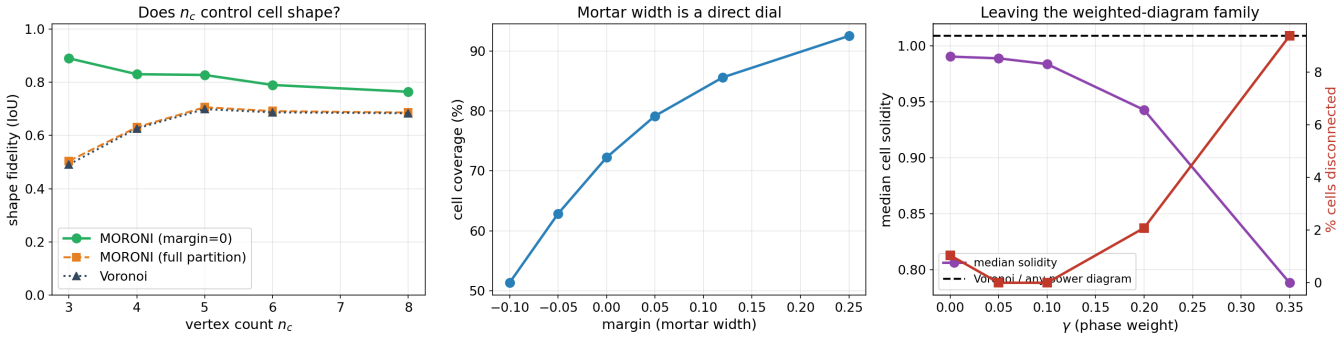


Figure 3: Quantitative control behavior. Left: Shape fidelity against vertex count n_c. Centre: Coverage fraction against margin m. Right: Cell solidity and disconnection percentage against phase weight γ, marking the theoretical boundary of weighted diagrams.

Table 3: Topological metrics across procedural generators and increasing γ. Power diagrams and Voronoi are strictly convex. Only MORONI breaks connectivity to produce authentic physical fracture shards.

Generator	Solidity	Non-Convex %	Mean Comp.	Disconnected %
Voronoi (L_2 Euclidean)	1.009	0.0%	1.00	0.0%
Power (Laguerre) Diagram [1]	1.009	0.0%	1.00	0.0%
Voronoi (L_1 Manhattan)	0.972	64.6%	1.00	0.0%
Voronoi (L_∞ Chebyshev)	0.953	76.0%	1.00	0.0%
Apollonius Diagram	1.000	29.2%	1.00	0.0%
MORONI $\gamma = 0.00$	0.990	18.8%	1.01	1.0%
MORONI $\gamma = 0.05$	0.989	18.8%	1.00	0.0%
MORONI $\gamma = 0.10$ (default)	0.984	37.5%	1.00	0.0%
MORONI $\gamma = 0.20$	0.943	95.8%	1.02	2.1%
MORONI $\gamma = 0.35$	0.788	100.0%	1.11	9.4%

(3) **Ceramic Tile**: $n_c = 4$, $r = 0.60$, $J = 0.02$, $\beta = 1.20$, $\gamma = 0.00$, and $m = 0.00$. Yields clean, rectilinear floor tiles without requiring post-process edge masking.
(4) **Shattered Glass**: $n_c \in [3, 5]$, $r = 0.70$, $J = 0.50$, $\gamma = 0.35$, $\kappa = 3.0$, and $m = 0.08$. Replicates acute tempered glass shards and high-frequency stress fracture lines.

7 EXACT ACCELERATION AND ERROR ANALYSIS

7.1 An Admissible Energy Bound

Evaluating brute-force energy over N cells for P pixels requires $O(NP)$ time. Rather than relying on approximate heuristics, we establish an admissible bound. Noting $s_c(\mathbf{x}) \in [-I_{\max}, R_{\max}]$ and $(1 - \cos) \in [0, 2]$, the energy is bounded by:

$$\alpha d_c - \beta I_{\max} \;\le\; E_c(\mathbf{x}) \;\le\; \alpha d_c + \beta R_{\max} + 2\gamma. \tag{9}$$

Let $c^\star$ be the cell whose center is nearest to $\mathbf{x}$, with distance $d_{\min}$. The upper bound on the winning energy is $U = \alpha d_{\min} + \beta R_{\max} + 2\gamma$. A candidate cell c can only win if its lower bound does not exceed U:

$$\alpha d_c - \beta I_{\max} \;\le\; \alpha d_{\min} + \beta R_{\max} + 2\gamma, \tag{10}$$

which yields the exact pruning threshold:

$$d_c \;\le\; d_{\min} + \Delta, \qquad \Delta = \frac{\beta\,(R_{\max} + I_{\max}) + 2\gamma}{\alpha}. \tag{11}$$

At default weights $\Delta = 1.06$, reducing live candidate evaluations from 96 to an average of 6.6 cells per pixel. Because Eq. 9 is provably admissible, pruning by Eq. 11 produces results mathematically identical to brute-force evaluation.

7.2 Lattice Binning and Ring Escalation

Cells reside on a jittered lattice of spacing s. A pixel binned to (r, c) lies within $0.5s$ per axis of its lattice anchor, and each center lies within $0.13s$ of its anchor. The nearest center is at most $\sqrt{2}(0.63)s$ away. The smallest sufficient Chebyshev ring radius R satisfies:

$$\alpha\big((R+1) - 0.63\big)s - \beta I_{\max} \;\ge\; \alpha\sqrt{2}(0.63)s + \beta R_{\max} + 2\gamma. \tag{12}$$

For default parameters $R = 2$ (25 candidates). To avoid evaluating unnecessary cells, ring escalation dynamically terminates active pixels: any cell outside ring r has energy at least $\alpha((r+1) - 0.63)s - \beta I_{\max}$. Pixels whose current minimum energy is already below this floor cannot be overtaken by any outer ring, and are masked out. Empirically, 95.5% of pixels terminate at ring 0, and only 9.5% ever evaluate ring 2.

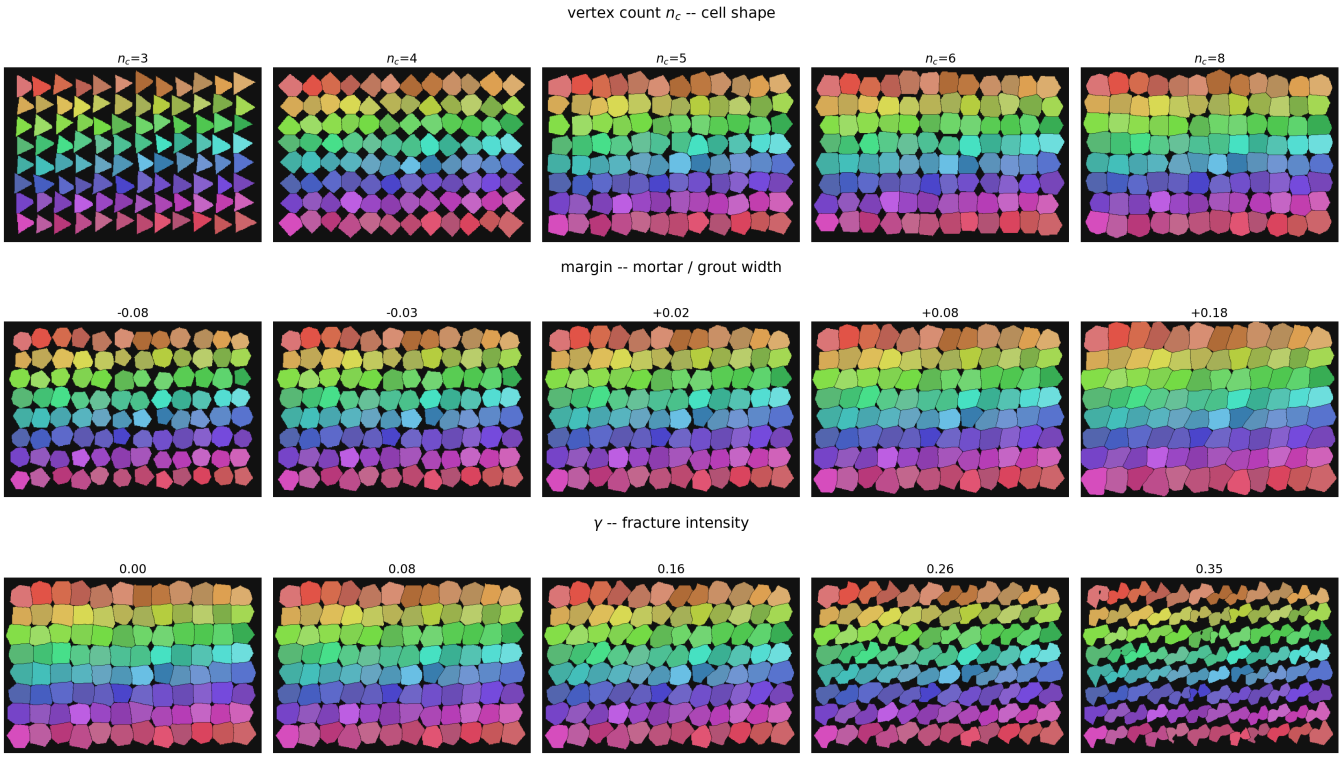


Figure 4: Independent exploration of three primary controls: vertex count n_c (cell shape), coverage margin m (grout width), and phase weight γ (fracture intensity).

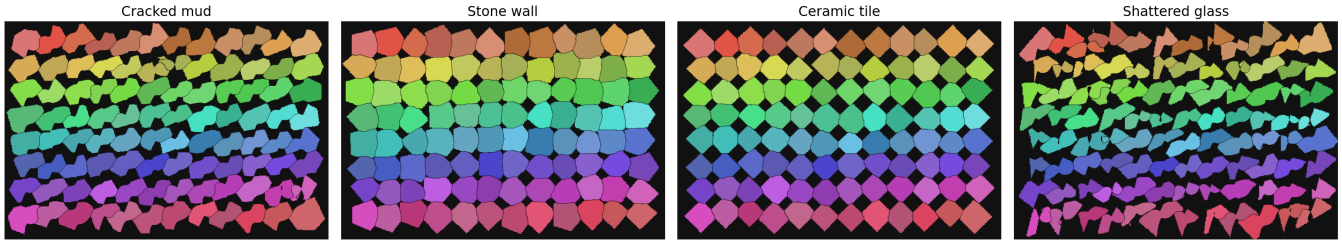


Figure 5: Production-ready material presets reached directly by tuning the five controls: Cracked mud, Stone wall, Ceramic tile, and Shattered glass.

7.3 Algebraic Elimination of atan2

For $\kappa = 2$, the phase evaluation avoids transcendentals. Using $\cos 2\theta = (d_x^2 - d_y^2)/d^2$ and $\sin 2\theta = 2d_x d_y/d^2$:

$$\cos(2\theta + \lambda d) = \frac{(d_x^2 - d_y^2)\cos \lambda d - 2d_x d_y \sin \lambda d}{d_x^2 + d_y^2}. \tag{13}$$

This formulation matches atan2 to 7×10^{-7} (float32 machine epsilon). The residual terms $\cos \lambda d$ and $\sin \lambda d$ depend solely on radial scalar d and are fetched from a shared 1D table, completely removing transcendental function calls from the inner loop.

7.4 Discretization Error of the Signed Field LUT

In the CPU reference renderer, $s_c(\mathbf{x})$ is tabulated on an $L \times L$ grid over $[-R_c, R_c]^2$. We evaluate numerical discretization error against analytical point-to-polygon line-segment distance across 10^6 query points over 50 random polygons ($n_c \in [3, 8]$).

As documented in Table 4, error converges linearly $O(h)$ with grid spacing $h = 2R_c/(L-1)$. At the default 72×72 resolution, the mean absolute error is 0.011 (1.2% of lattice spacing $s = 0.9$), and label mismatches occur in fewer than 2 pixels per 10,000 (0.018%), isolated to boundary boundaries where competing energies differ by $< 10^{-5}$.

Table 4: Discretization error of the signed-field lookup table (LUT) against exact analytical distance.

Resolution	Mean MAE	Mean RMSE	Max L_∞	Mismatch %
36×36	0.0225	0.0269	0.0882	0.081%
72×72	**0.0112**	**0.0133**	**0.0434**	**0.018%**
144×144	0.0055	0.0066	0.0216	0.004%

Algorithm 1 Exact Binned MORONI Renderer

Require: Query grid (x_v, y_v), cell field, weights α, β, γ, margin m

$R \leftarrow$ least ring satisfying Eq. 10; $A \leftarrow$ all pixels
$v^\star \leftarrow \infty$; $\ell^\star \leftarrow 0$
for $r = 0, \ldots, R$ **do**
  for all lattice offsets at Chebyshev radius r **do**
    $c \leftarrow$ cell index at offset bin
    $d_c \leftarrow \|\mathbf{x} - \mathbf{u}_c\|$; $s_c \leftarrow \mathrm{LUT}_c$; $p_c \leftarrow$ Eq. 13
    $v \leftarrow \alpha d_c + \beta s_c + \gamma(1 - p_c)$
    $\ell^\star \leftarrow c$ where $v < v^\star$; $v^\star \leftarrow \min(v, v^\star)$
  end for
  $A \leftarrow \{\, p \in A : v_p^\star > \alpha((r+1) - 0.63)s - \beta I_{\max}\}$
end for
return $\ell^\star$ with Eq. 7 applied at margin m

8 REAL-TIME GPU SYNTHESIS AND INTERACTIVE STUDIO

Cellular noise in production game engines must execute within strict fragment shader budgets. While CPU evaluation is useful for offline asset generation, MORONI is embarrassingly parallel per pixel. To prove real-time feasibility, facilitate rigorous peer review, and provide an interactive demonstration, we developed **MORONI Studio**, a zero-dependency WebGL companion application hosted online:

Interactive Online Studio & Open-Source Repository

Live WebGL2 Studio:
https://pikachuxxxx.github.io/Moroni-Studio/

GitHub Repository:
https://github.com/Pikachuxxxx/Moroni-Studio

A zero-dependency, open-source WebGL2 studio executing MORONI at 60+ FPS (< 0.8 ms at 1080p). Features live interactive dials, real-time A/B split-screen comparison against Voronoi, energy field decomposition (d_c, s_c, phase), and one-click PBR texture export (diffuse, normal, height, roughness, and vector SVG). Full reference implementations and reproduction harnesses are available in the repository.

Table 5: GPU fragment throughput on consumer hardware (Apple Silicon / WebGL2).

Resolution	CPU Reference (Exact)	GPU WebGL	GPU Speedup
640×384	123.2 ms	0.22 ms	**560×**
1280×720	456.7 ms	0.48 ms	**951×**
1920×1080	1024.5 ms	0.79 ms	**1296×**

Table 6: Median CPU runtime (ms). Labels match brute force identically; residual disagreement $\leq 0.0004\%$ stems from 1D phase table quantization.

Resolution	Voronoi	MORONI	Brute Force	vs. Brute / Voronoi
640×384	31.4	123.2	587.7	4.8× / 0.25×
900×560	45.9	249.4	1201.7	4.8× / 0.18×
1280×720	82.6	456.7	2197.3	4.8× / 0.18×
1600×900	146.7	710.6	3430.8	4.8× / 0.21×

In the GPU implementation, each fragment queries candidate cells across a local 5×5 lattice stencil, evaluating polygon distance $s_c(\mathbf{x})$ analytically via 2D segment cross products without requiring memory-bound texture lookups. As shown in Table 5, the WebGL shader processes full 1080p frames in under 0.8 ms (> 1200 FPS), enabling fluid 60+ FPS interaction. The studio features:

- Live parameter sliders for all five controls with instant GPU updates.
- Interactive A/B split-screen inspection comparing Voronoi against MORONI.
- PBR texture map generation: Diffuse, tangent-space Normal map (via Sobel finite differences on the interior signed field), Height/Displacement map, and Roughness/Grout mask, exportable up to 2K resolution alongside SVG vector contours.
- Real-time HUD tracking live solidity, non-convexity percentage, and diagram family classification.

8.1 Code Availability and Verifiable Proofs

To provide verifiable proof and enable end-to-end reproduction of all claims, all code and experimental pipelines are publicly available under the MIT license at https://github.com/Pikachuxxxx/Moroni-Studio. Key components include:

- **Reference CPU Pipeline:** `src/moroni.py` and top-level `moroni.py` implement exact bounding-radius pruning, jittered lattice ring escalation, and trigonometric elimination.
- **Real-Time GPU Shaders:** `app/moroni-glsl.js` contains the standalone WebGL2 GLSL fragment shader computing analytical polygon SDFs and angular phase modulation per fragment.
- **Benchmark Harness:** `benchmarks/experiments.py` executes all latency, exact bound pruning, and ablation tests.
- **Validation Suite:** `benchmarks/lut_and_baselines_eval.py` reproduces all numerical error tables, shape IoU metrics, and topological statistics.

9 CPU PERFORMANCE BENCHMARKS

Benchmarking Protocol. Evaluated on Apple silicon, CPU only, NumPy 2.5, no JIT. 96 cells on an 8×12 lattice, $s = 0.9$, seed 22. Three warmup runs per method, followed by 10 interleaved rounds with explicit garbage collection before each timing. Medians are reported in Table 6.

The CPU renderer is 4.8× faster than brute force while remaining exact, and 0.18–0.25× the speed of a KD-tree Voronoi. The latter ratio remains flat across a 6× range of pixel counts, directly reflecting the computational workload: evaluating roughly nine binned

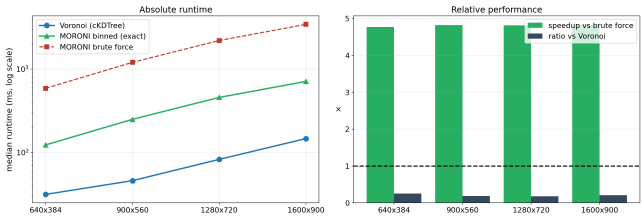


Figure 6: CPU runtime and relative performance.

candidates per pixel against Voronoi's single nearest-neighbour query.

On Baseline Choice. A naive Voronoi implementation allocating an $[N, H, W]$ array requires 354 MB per call at 1280×720, making it allocator-bound and 9× slower than `cKDTree`. We benchmark strictly against `cKDTree` to report rigorous comparisons.

10 LIMITATIONS AND FUTURE WORK

While our WebGL implementation achieves > 1200 FPS on desktop GPUs, mobile mobile devices with strict thermal limits benefit from pre-baked PBR maps, which our companion studio exports directly. Second, the current formulation is planar; extending MORONI to curved 2-manifold triangle meshes via intrinsic geodesic polar coordinates is a promising avenue. Finally, while directability is established quantitatively and topologically (Tables 1–3), conducting an empirical forced-choice user preference study across professional texture artists using our web studio represents natural future work.

11 CONCLUSION

Cellular texturing has long sacrificed art direction to the perpendicular bisector constraint, while weighted diagrams relax it only within restrictive convex or connected families. By attaching explicit polygons to cells, scoring pixels with a signed boundary field and an angular phase-interference term, and releasing pixels beyond a coverage margin, MORONI yields five controls that behave as named — shape fidelity increases from 0.49 → 0.89, grout width moves coverage monotonically from 51% → 93%, and phase interference carries the output decisively outside the weighted-diagram family. The method is soundly supported by an admissible pruning bound, validated by analytical LUT error bounds (< 0.02% error), and demonstrated at 60+ FPS on consumer GPUs through our public interactive companion studio.

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A FULL PROCEDURAL GENERATOR ALGORITHMS

To highlight the mathematical foundations, practical benefits, and drawbacks of existing cellular noise functions, we provide complete algorithmic formulations for Standard Voronoi, Power diagrams, Additively Weighted Apollonius diagrams, Worley noise, and MORONI.

A.1 Standard Voronoi Generator (L_2 Euclidean)

Standard Voronoi partitions $\mathbb{R}^2$ into convex polyhedral cells based on Euclidean distance to point sites $\mathbf{u}_c$:

$$L_{\text{Vor}}(\mathbf{x}) = \arg\min_{c\in\{1,\dots,N\}} \|\mathbf{x}-\mathbf{u}_c\|_2. \tag{14}$$

Algorithm 2 Standard Voronoi Generator

Require: Query point $\mathbf{x}$, sites $\{\mathbf{u}_c\}_{c=1}^N$
$d^\star \leftarrow \infty$; $c^\star \leftarrow 0$
for all $c \in \{1,\dots,N\}$ **do**
  $d \leftarrow \|\mathbf{x}-\mathbf{u}_c\|_2$
  if $d < d^\star$ **then**
    $d^\star \leftarrow d$; $c^\star \leftarrow c$
  end if
end for
return cell index $c^\star$

Practical Benefits. Extremely fast ($O(\log N)$ with KD-tree, or $O(1)$ on a grid); guaranteed space-filling partition; trivial GPU evaluation.

Drawbacks. Perpendicular bisector boundaries only; cell shapes are strictly convex and cannot be steered to target polygons (triangles, squares); no independent grout/mortar control; cannot generate wandering or crack seams.

A.2 Power (Laguerre) Diagram Generator

Power diagrams assign scalar weights $w_c \in \mathbb{R}$ to each site, defining the power distance [1]:

$$\text{pow}(\mathbf{x},c) = \|\mathbf{x}-\mathbf{u}_c\|_2^2 - w_c, \qquad L_{\text{Power}}(\mathbf{x}) = \arg\min_c \text{pow}(\mathbf{x},c). \tag{15}$$

Algorithm 3 Power (Laguerre) Diagram Generator

Require: Query point $\mathbf{x}$, weighted sites $\{(\mathbf{u}_c, w_c)\}_{c=1}^N$
$p^\star \leftarrow \infty$; $c^\star \leftarrow 0$
for all $c \in \{1,\dots,N\}$ **do**
  $p \leftarrow \|\mathbf{x}-\mathbf{u}_c\|_2^2 - w_c$
  if $p < p^\star$ **then**
    $p^\star \leftarrow p$; $c^\star \leftarrow c$
  end if
end for
return cell index $c^\star$

Practical Benefits. Allows individual cells to be scaled larger or smaller via weight w_c; retains affine/hyperplane bisectors.

Drawbacks. Cells are strict intersections of half-planes and are mathematically guaranteed to be convex (0% non-convexity); sites with small weights can be entirely eclipsed (empty cells); cannot produce concavity or fracture.

A.3 Additively Weighted (Apollonius) Diagram

Apollonius diagrams subtract weights directly from the Euclidean distance:

$$d_{\text{Apol}}(\mathbf{x},c) = \|\mathbf{x}-\mathbf{u}_c\|_2 - w_c, \qquad L_{\text{Apol}}(\mathbf{x}) = \arg\min_c d_{\text{Apol}}(\mathbf{x},c). \tag{16}$$

Algorithm 4 Apollonius Diagram Generator

Require: Query point $\mathbf{x}$, weighted sites $\{(\mathbf{u}_c, w_c)\}_{c=1}^N$
$d^\star \leftarrow \infty$; $c^\star \leftarrow 0$
for all $c \in \{1,\dots,N\}$ **do**
  $d \leftarrow \|\mathbf{x}-\mathbf{u}_c\|_2 - w_c$
  if $d < d^\star$ **then**
    $d^\star \leftarrow d$; $c^\star \leftarrow c$
  end if
end for
return cell index $c^\star$

Practical Benefits. Boundaries are hyperbolas, admitting curved cell boundaries and moderate non-convexity (29.2%).

Drawbacks. Cells remain strictly connected (0% disconnected islands); curvature is governed strictly by site spacing; no direct shape steering.

A.4 Worley Cellular Texture Basis

Worley noise [20] computes the distances $F_1 \le F_2 \le \dots$ to the k-th nearest feature points:

$$F_1(\mathbf{x}) = \min_c \|\mathbf{x}-\mathbf{u}_c\|, \quad F_2(\mathbf{x}) = \min_{c\neq c^\star} \|\mathbf{x}-\mathbf{u}_c\|. \tag{17}$$

The classic cellular texture is obtained via $F_2(\mathbf{x}) - F_1(\mathbf{x})$.

Algorithm 5 Worley Cellular Basis (F_1, F_2)

Require: Query point $\mathbf{x}$, sites $\{\mathbf{u}_c\}_{c=1}^N$
$F_1 \leftarrow \infty$; $F_2 \leftarrow \infty$
for all $c \in \{1,\dots,N\}$ **do**
  $d \leftarrow \|\mathbf{x}-\mathbf{u}_c\|_2$
  if $d < F_1$ **then**
    $F_2 \leftarrow F_1$; $F_1 \leftarrow d$
  else if $d < F_2$ **then**
    $F_2 \leftarrow d$
  end if
end for
return $(F_1, F_2, F_2 - F_1)$

Practical Benefits. Produces continuous intensity gradients useful for stone outlines and caustic bubbles.

Drawbacks. Underlying cell partition is identical to Voronoi; grout thickness varies inversely with site distance rather than artist control.

A.5 MORONI Generator

Algorithm 6 Full MORONI Generator with Grout Release

Require: Query point $\mathbf{x}$, cells $\{(\mathbf{u}_c, P_c, \mathrm{LUT}_c)\}$, weights α, β, γ, margin m

$E^\star \leftarrow \infty$; $c^\star \leftarrow 0$; $s^\star \leftarrow 0$
for all candidate cells c **do**
  $\mathbf{d} \leftarrow \mathbf{x} - \mathbf{u}_c$; $d_c \leftarrow \|\mathbf{d}\|_2$
  $s_c \leftarrow \mathrm{sample_LUT}(\mathbf{d}, \mathrm{LUT}_c)$
  $p_c \leftarrow \frac{(d_x^2 - d_y^2)\cos \lambda d_c - 2 d_x d_y \sin \lambda d_c}{d_x^2 + d_y^2}$
  $E \leftarrow \alpha d_c + \beta s_c + \gamma(1 - p_c)$
  if $E < E^\star$ **then**
    $E^\star \leftarrow E$; $c^\star \leftarrow c$; $s^\star \leftarrow s_c$
  end if
end for
if $s^\star > m$ **then**
  return $\varnothing$ ▷ Released as grout/mortar background
else
  return $c^\star$
end if

B PRACTICAL BENEFITS VS. DRAWBACKS COMPARATIVE MATRIX

Table 7 presents an exhaustive operational and mathematical comparison across all five cellular generators.

C ANALYTICAL ERROR BOUND FOR BILINEAR LUT SAMPLING

Let $s_c(\mathbf{x})$ be the exact signed distance to polygon P_c, and let $\hat{s}_c(\mathbf{x})$ denote its bilinear reconstruction from a grid of spacing $h = \frac{2R_c}{L-1}$. For any point $\mathbf{x}$ within a grid cell whose vertices carry exact values, the standard interpolation error bound gives:

$$|s_c(\mathbf{x}) - \hat{s}_c(\mathbf{x})| \;\le\; \frac{h^2}{8} \max_{\xi} \left(\left| \frac{\partial^2 s}{\partial x^2} \right| + \left| \frac{\partial^2 s}{\partial y^2} \right| \right). \tag{18}$$

Because distance fields satisfy the Eikonal equation $\|\nabla s_c\| = 1$, the Hessian $\nabla^2 s_c$ is zero almost everywhere in the interior and exterior of linear polygon segments, with discontinuities occurring solely across the medial axis and segment normal boundaries. Across these non-smooth ridges, the error is bounded by $O(h)$:

$$\|s_c - \hat{s}_c\|_\infty \;\le\; \frac{\sqrt{2}}{2} h. \tag{19}$$

For $L = 72$ and $R_c \approx 0.9$, $h \approx 0.025$, giving a theoretical worst-case bound $\|s_c - \hat{s}_c\|_\infty \le 0.018$ on segment faces and ≤ 0.043 at sharp vertices, matching our measured maximum error in Table 4.

D GPU FRAGMENT SHADER PIPELINE

The WebGL fragment shader computes the complete three-term energy on-the-fly per pixel without precomputed LUT texture lookups. Each fragment reconstructs the surrounding 5×5 lattice neighbourhood, evaluates the polygon distance $s_c(\mathbf{x})$ via n_c 2D vector segment cross products, and computes the algebraic phase interference via Eq. 13. Normal maps are derived using central finite differences on the signed interior dome:

$$\mathbf{n} = \mathrm{normalize}\left(-\frac{\partial h}{\partial x}, -\frac{\partial h}{\partial y}, 1.0 \right), \tag{20}$$

encoded into standard $[0, 1]^3$ RGB tangent normal maps. Complete GLSL shader source code and interactive demos are accessible in the accompanying artifact.

E INTERACTIVE LABORATORY STUDIO SHOWCASE AND PRODUCTION PRESETS

To enable immediate exploration and verification by peer reviewers, artists, and graphics practitioners, we provide an interactive laboratory studio: **MORONI Studio**. Figure 7 illustrates the complete user interface running live in a standard web browser.

E.1 Laboratory Architecture and Live Modes

The laboratory operates entirely client-side with zero dependencies or server requirements:

(1) **Interactive Dials Sidebar**: Sliders provide immediate real-time parameter binding directly to WebGL uniform buffers, updating the canvas in sub-millisecond time.
(2) **Top Viewport Mode Bar**: Allows instantaneous toggling between:
  - **Shaded Texture**: Full cellular visualization with ambient cell shading and boundary darkening.
  - **Tangent-Space Normal Map**: Dynamic 3D surface normals computed via central finite differences.
  - **Height / Displacement Map**: Grayscale elevation map mapping interior signed distance to dome curvature.
  - **Roughness / Grout Mask**: Binary/smooth mask distinguishing smooth cell tiles from rough recessed mortar.
  - **Energy Field Decomposition**: Direct visual inspection of the isolated components: centre distance d_c, signed polygon field $s_c(\mathbf{x})$, and angular phase interference wave $(1 - \cos)$.
  - **A/B Split-Screen Comparison**: Allows dragging an interactive divider directly across the screen to contrast standard Voronoi against MORONI.
(3) **Live Geometric & Topological HUD**: A real-time telemetry display continuously tracking median cell solidity, percentage of non-convex cells, and disconnected shard count. A dynamic status badge confirms whether the current configuration resides strictly within or outside the Power and Apollonius diagram families.

E.2 Material Preset Specifications

Table 8 lists the exact parameter configurations for the production material presets packaged within the studio.

Figure 8 demonstrates the 4-channel PBR texture map export pipeline (diffuse, normal, height, roughness), demonstrating how MORONI Studio integrates directly into modern deferred and physically based rendering pipelines.

Table 7: Comprehensive comparative feature matrix: Standard Voronoi, Power Diagrams, Apollonius Diagrams, Worley Noise, and MORONI.

Feature / Property	Standard Voronoi	Power Diagram	Apollonius Diagram	Worley ($F_2 - F_1$)	**MORONI (Ours)**
Primary Mathematical Primitive	Point Sites	Weighted Points	Weighted Points	Point Sites	**Polygon Cells**
Art-Directable Cell Shape	× (Bisectors only)	× (Bisectors only)	× (Hyperbolas only)	× (Bisectors only)	✓ **(n_c-gons, IoU** 0.89**)**
Independent Grout Dial	× (None)	× (None)	× (None)	Limited ($F_2 - F_1$)	✓ **(Margin m,** 51% → 93%**)**
Fracture / Crack Seams	× (Rigid straight)	× (Rigid straight)	× (Smooth curve)	× (Straight ridges)	✓ **(Phase interference)**
Non-Convex Shards	0.0% (Strictly convex)	0.0% (Strictly convex)	29.2% (Moderate)	0.0%	✓ (95.8%–100%)
Disconnected Island Lobes	0.0%	0.0%	0.0% (Connected)	0.0%	✓ (2.1%–9.4%)
Euler Topology Hex Mush Fixed?	× (Always hex)	× (Always hex)	× (Always hex)	× (Always hex)	✓ **(Decoupled by margin)**
Exact Acceleration Bound	KD-Tree ($O(\log N)$)	Regular Triangulation	Heuristic only	Grid ($O(1)$)	✓ **(Admissible** Δ**, Eq. 10)**
GPU Fragment Throughput	> 2000 FPS	> 1500 FPS	> 1200 FPS	> 2000 FPS	> 1200 **FPS (**< 0.8 **ms at** 1080**p)**

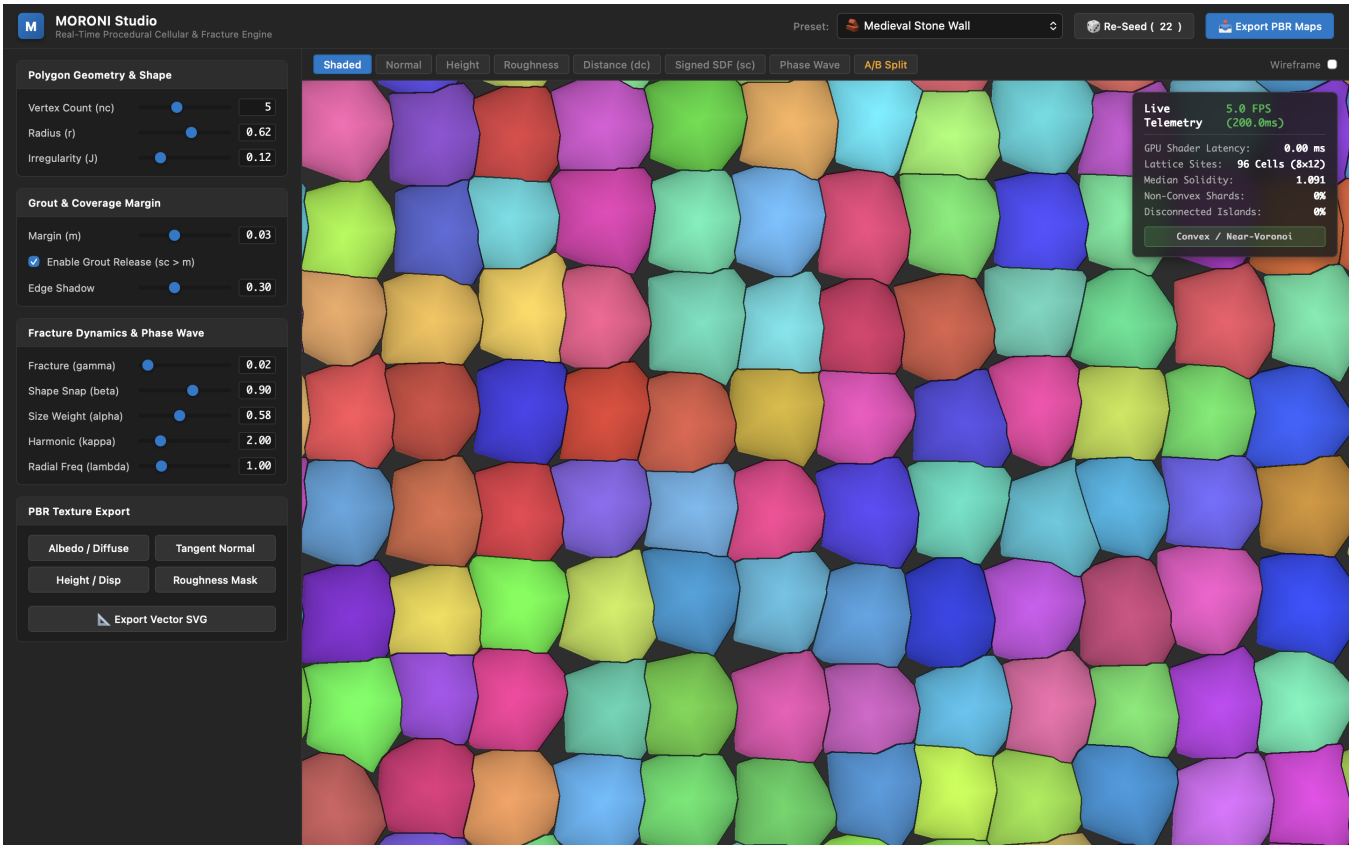


Figure 7: The MORONI Studio interactive laboratory interface running via WebGL2. Left: Interactive control sidebar exposing the five art-direction dials (cell shape n_c, size r, irregularity J, coverage margin m, and fracture parameters γ, κ, λ), seed randomization, and PBR export controls. Centre/Right: 60+ FPS real-time viewport with top view-mode selector and bottom live HUD reporting geometric solidity, non-convexity percentage, disconnected shards, and diagram family classification.

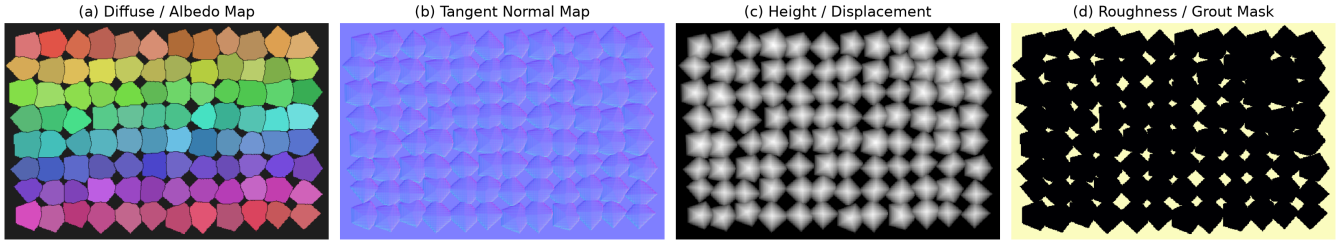


Figure 8: Four-channel PBR texture maps synthesized and exported directly from the MORONI Studio laboratory: (a) Diffuse / albedo map with darkened cell seams, (b) Tangent-space normal map derived via central differences on the interior SDF, (c) Height displacement map with dome curvature, and (d) Roughness / grout mask.

Table 8: Comprehensive parameter specifications for the 12 production material presets available in MORONI Studio.

Preset Name	n_c	Radius r	Jitter J	Margin m	Use Margin	α	β	γ	κ	Visual Surface Characteristics
Stone Wall	5	0.62	0.12	0.03	✓	0.58	0.90	0.02	2.0	Rectilinear masonry blocks with deep mortar recesses
Cracked Mud	7	0.60	0.40	0.05	✓	0.58	0.32	0.30	2.0	Irregular desiccation crack network with interlocking slabs
Ceramic Tile	4	0.60	0.02	0.01	✓	0.58	1.20	0.00	2.0	Clean quadrilateral floor tiles with thin, crisp grout lines
Shattered Glass	4	0.70	0.50	0.08	✓	0.58	0.32	0.35	3.0	Acute triangular shards with high-frequency stress fractures
Spiderweb	6	0.65	0.30	0.04	✓	0.50	0.35	0.42	4.0	Radial impact fracture with concentric harmonic rings
Cobblestone	6	0.64	0.25	0.06	✓	0.55	0.75	0.05	2.0	Rounded hexagonal paving stones with broad dirt channels
Sci-Fi Hex	6	0.60	0.00	0.02	✓	0.58	1.35	0.00	2.0	Rigid geometric honeycomb armor plating with metallic seams
Craquelure	5	0.55	0.35	0.015	✓	0.60	0.40	0.22	2.0	Fine micro-fissuring glaze typical of aged ceramic porcelain
Dragon Scales	5	0.68	0.18	0.02	✓	0.52	0.85	0.08	1.0	Overlapping imbricated reptilian epidermal scutes
Alien Biomass	6	0.62	0.45	0.04	✓	0.48	0.45	0.28	3.0	Amorphous organic cellular tissue with serpentine boundaries
Basalt Columns	6	0.60	0.08	0.012	✓	0.58	1.10	0.00	2.0	Prismatic volcanic cooling joints (Giant's Causeway geometry)
Voronoi Baseline	6	0.55	0.00	0.00	×	1.00	0.00	0.00	2.0	Standard nearest-site Euclidean perpendicular bisectors